Individual Assignment Instructions

Responsible AI

Principles:

1. **Mandatory AI usage disclosure**: You **MUST** include a statement at the end of every assignment explaining how and why you used generative AI (e.g., ChatGPT, Gemini, etc.).
2. **Honesty and transparency**: The statement must be detailed and truthful, clearly explaining the role of AI in your work. For example: *"I used AI to draft the project report outline"* or *"AI was used to refine the report structure"* rather than a vague *"I used AI."*
3. **Full responsibility**: The content you submit is entirely your responsibility. You must verify, understand, and be able to defend all information produced with AI.

Penalties:

Failure to comply with these rules, including omission of the AI usage statement or providing a dishonest statement, will be treated as **plagiarism** or academic misconduct. Penalties may include, but are not limited to:

- Provide verbal or written warnings.
- Assignment downgrade up to zero (0).
- Failure of the course (grade = E) or other academic sanctions in accordance with university policies.

Assignment 3 (Week 7): Community Analysis and Visualization

Deadline: Week 9

Objective: to implement SNA using small public datasets and visualize the community

Instructions:

- Use a dataset from official repositories, different for each student (e.g., SNAP, KONECT), with fewer than 500 nodes.
- Perform community detection with the **Louvain**, **Label Propagation, and Girvan Newman** algorithms.

Compare the results:

- Differences in the number of communities
- Nodes that change communities

Create visualizations for each algorithm:

- Distinct colors for each community
- Visible node labels

Write a report contain an interpretation of:

- How many communities are detected?
- Does the community division make sense given the dataset context?

Attach the original dataset file and the analysis results.

Responsible AI (NO AI)